



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Code & Data



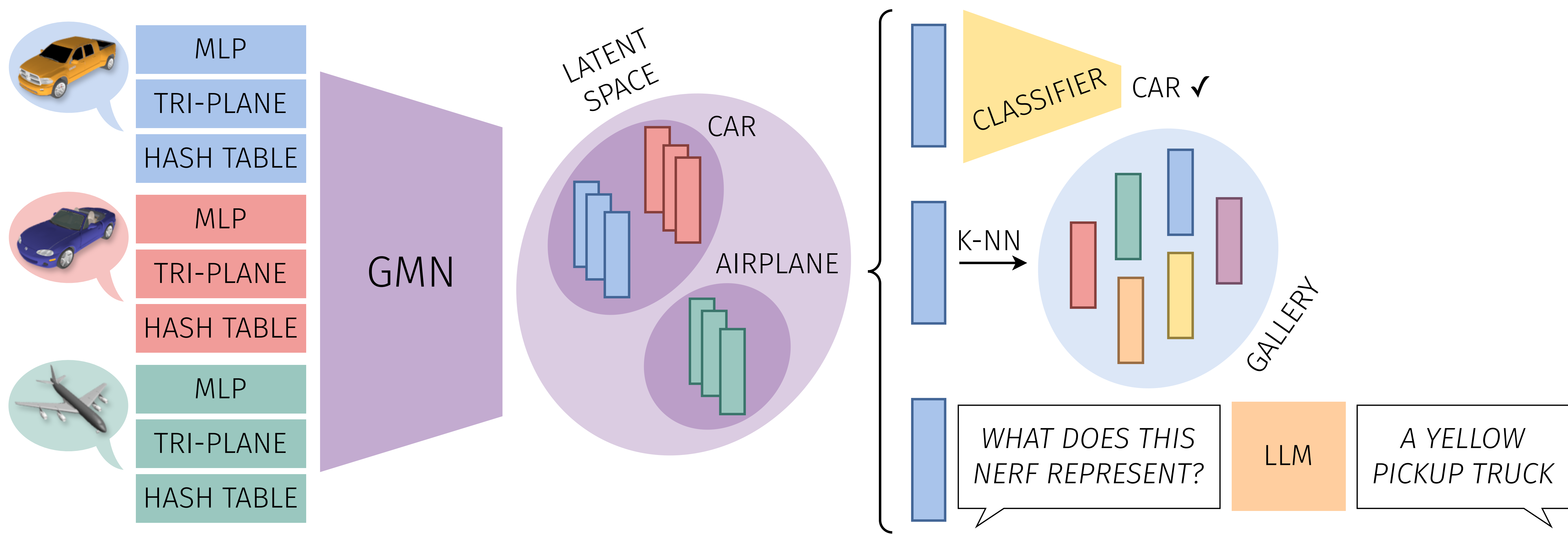
Weight Space Representation Learning on Diverse NeRF Architectures

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Learning Representations

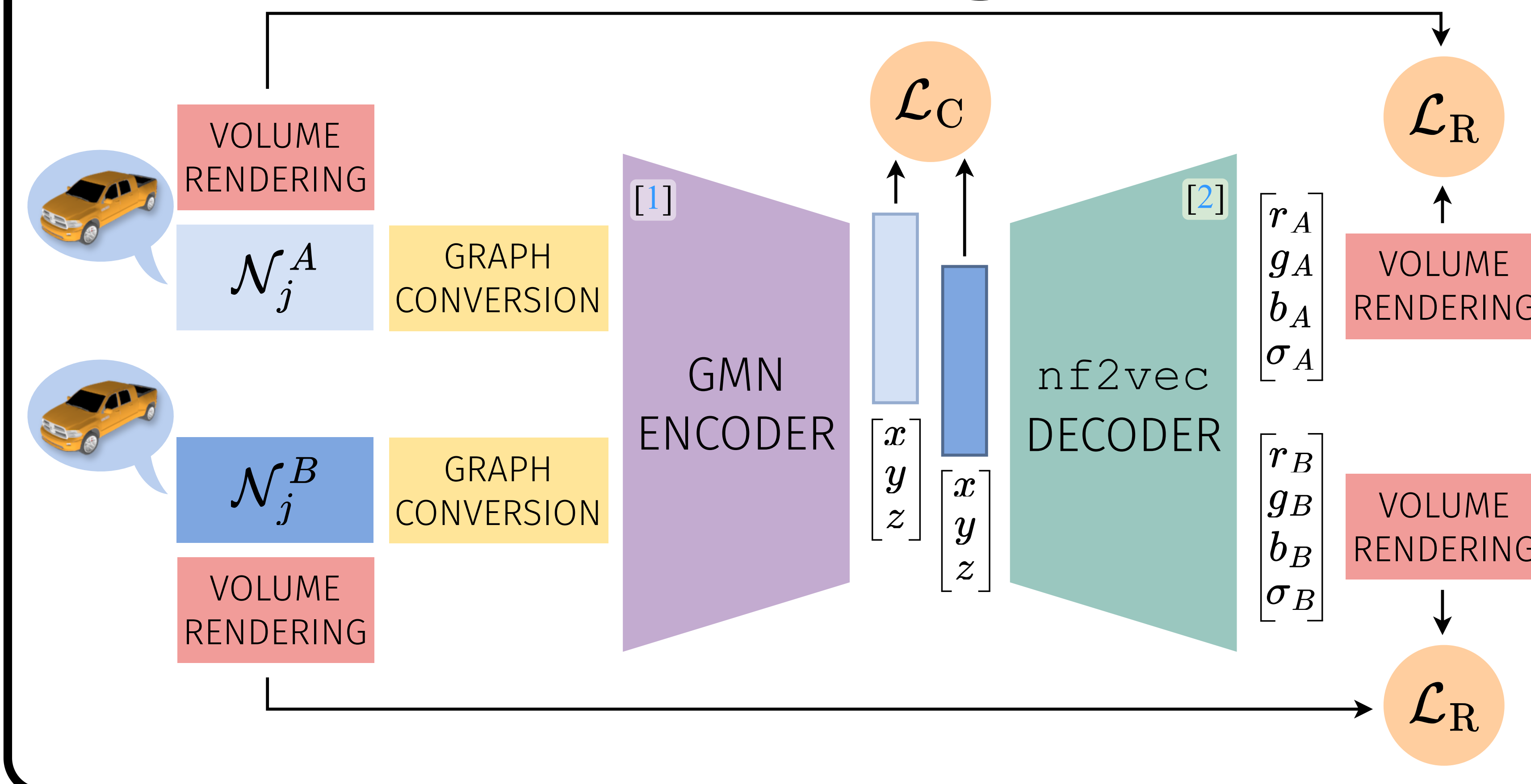
Overview



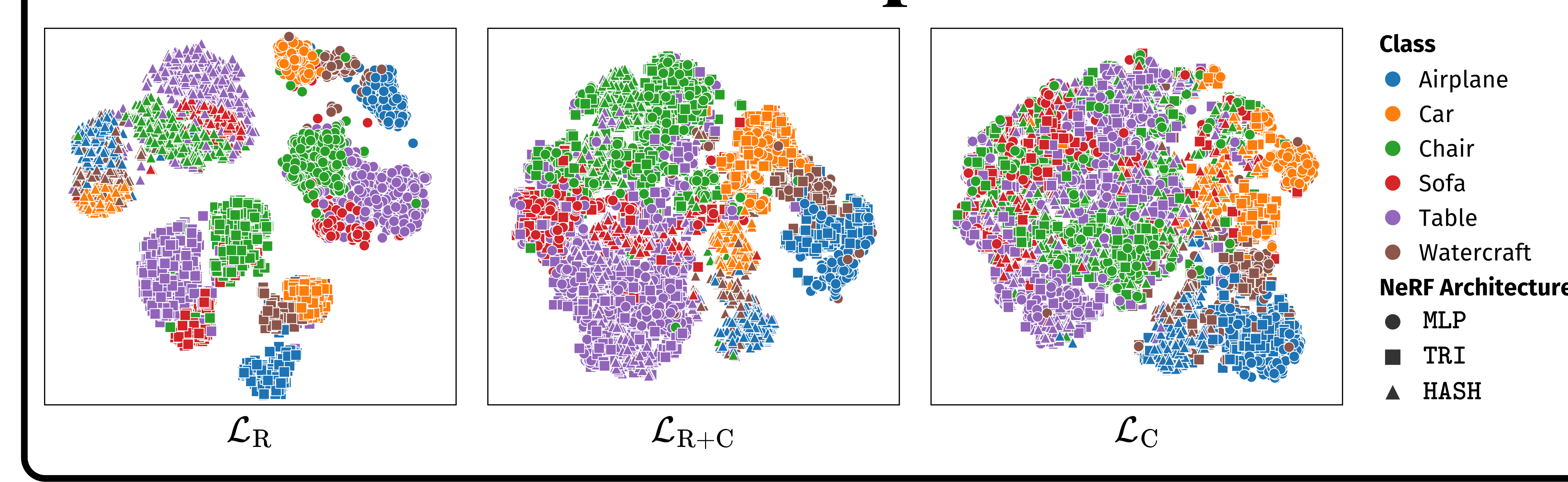
Our framework:

- Performs tasks on NeRFs parameterized by **diverse architectures** by processing their **weights**
- Leverages a **contrastive learning** objective to create a latent space where NeRFs with similar content are close to each other, regardless of their architecture
- Is the first to tackle downstream tasks on NeRFs parameterized by **multi-resolution hash tables**
- Can process **unseen** NeRF **architectures** and generalize to NeRFs trained on **unseen data**
- Achieves comparable or superior results to previous methods operating on single architectures

Training



Latent Space



Retrieval

Recall@k (%) ↑							
Method	k	MLP/TRI	MLP/HASH	TRI/MLP	TRI/HASH	HASH/MLP	HASH/TRI
nf2vec [2]							
Cardace et al. [3]	—	—	—	—	—	—	—
RANDOM	1	0.03	0.03	0.03	0.03	0.03	0.03
$\mathcal{L}_R$ (ours)	1	1.80	0.43	4.48	1.67	1.06	0.38
$\mathcal{L}_{R+C}$ (ours)	1	30.62	14.77	33.27	13.17	13.45	9.46
RANDOM	5	0.13	0.13	0.13	0.13	0.13	0.13
$\mathcal{L}_R$ (ours)	5	5.28	1.82	13.30	6.12	3.84	1.37
$\mathcal{L}_{R+C}$ (ours)	5	59.37	37.98	61.24	33.40	32.09	25.13
RANDOM	10	0.25	0.25	0.25	0.25	0.25	0.25
$\mathcal{L}_R$ (ours)	10	8.19	3.08	19.24	9.23	6.09	2.68
$\mathcal{L}_{R+C}$ (ours)	10	72.67	51.25	72.62	45.01	42.83	36.31

Recall@k (%) ↑							
Method	k	MLP ^{OB} /TRI ^{OB}	MLP ^{OB} /HASH ^{OB}	TRI ^{OB} /MLP ^{OB}	TRI ^{OB} /HASH ^{OB}	HASH ^{OB} /MLP ^{OB}	HASH ^{OB} /TRI ^{OB}
nf2vec [2]							
Cardace et al. [3]	—	—	—	—	—	—	—
RANDOM	1	0.05	0.05	0.05	0.05	0.05	0.05
$\mathcal{L}_R$ (ours)	1	2.60	0.48	2.71	0.48	0.74	0.32
$\mathcal{L}_{R+C}$ (ours)	1	9.82	6.10	8.33	1.75	6.74	2.92
RANDOM	5	0.27	0.27	0.27	0.27	0.27	0.27
$\mathcal{L}_R$ (ours)	5	6.37	1.11	7.27	1.91	2.92	0.69
$\mathcal{L}_{R+C}$ (ours)	5	23.67	17.94	20.70	5.73	19.59	8.49
RANDOM	10	0.53	0.53	0.53	0.53	0.53	0.53
$\mathcal{L}_R$ (ours)	10	9.66	2.39	10.99	3.24	4.88	1.80
$\mathcal{L}_{R+C}$ (ours)	10	33.65	26.38	27.55	9.50	28.13	12.74

MLP ^{OB} / MLP					
QUERY	1-NN	2-NN	3-NN	4-NN	5-NN

MLP ^{OB} / TRI					
QUERY	1-NN	2-NN	3-NN	4-NN	5-NN

MLP ^{OB} / HASH					
QUERY	1-NN	2-NN	3-NN	4-NN	5-NN

Classification

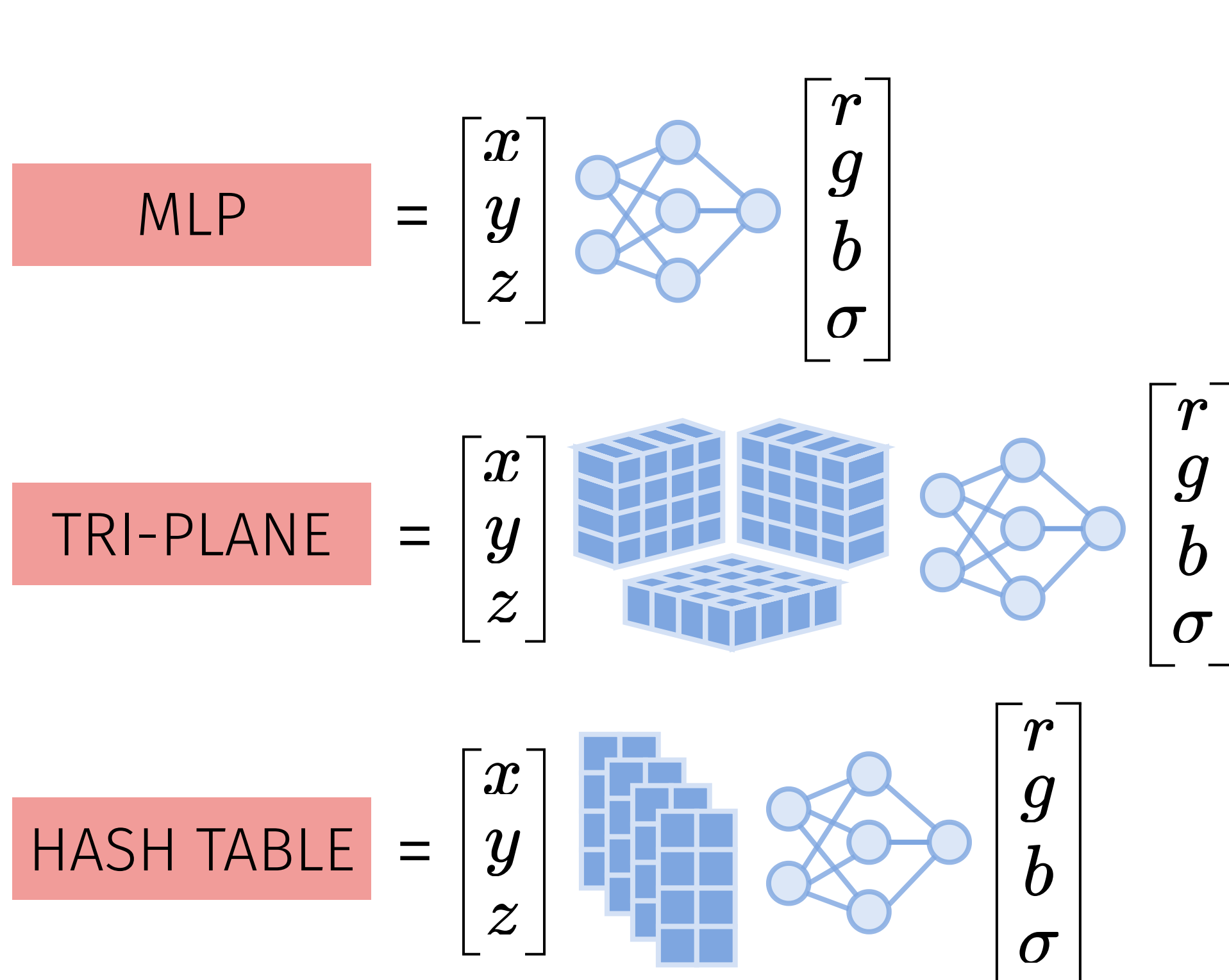
Method	Classifier Training Set	MLP	TRI	HASH
Accuracy (%) ↑				
nf2vec [2]	—	—	—	—
Cardace et al. [3]	—	—	—	—
$\mathcal{L}_R$ (ours)	ALL	93.6	94.0	92.4
$\mathcal{L}_{R+C}$ (ours)	ALL	90.7	90.6	90.0
$\mathcal{L}_R$ (ours)	MLP	93.8	25.3	19.3
$\mathcal{L}_{R+C}$ (ours)	MLP	91.5	58.6	56.6
$\mathcal{L}_R$ (ours)	TRI	11.4	93.8	9.3
$\mathcal{L}_{R+C}$ (ours)	TRI	77.8	91.0	66.5
$\mathcal{L}_R$ (ours)	HASH	13.8	35.7	92.7
$\mathcal{L}_{R+C}$ (ours)	HASH	54.1	35.5	90.6

Method	Encoder Training Set	Accuracy (%) ↑
nf2vec [2]	—	92.1
Cardace et al. [3]	MLP	—
$\mathcal{L}_R$ (ours)	MLP	93.6
nf2vec [2]	—	—
Cardace et al. [3]	TRI	93.1
$\mathcal{L}_R$ (ours)	TRI	94.0
nf2vec [2]	—	—
Cardace et al. [3]	HASH	—
$\mathcal{L}_R$ (ours)	HASH	92.5

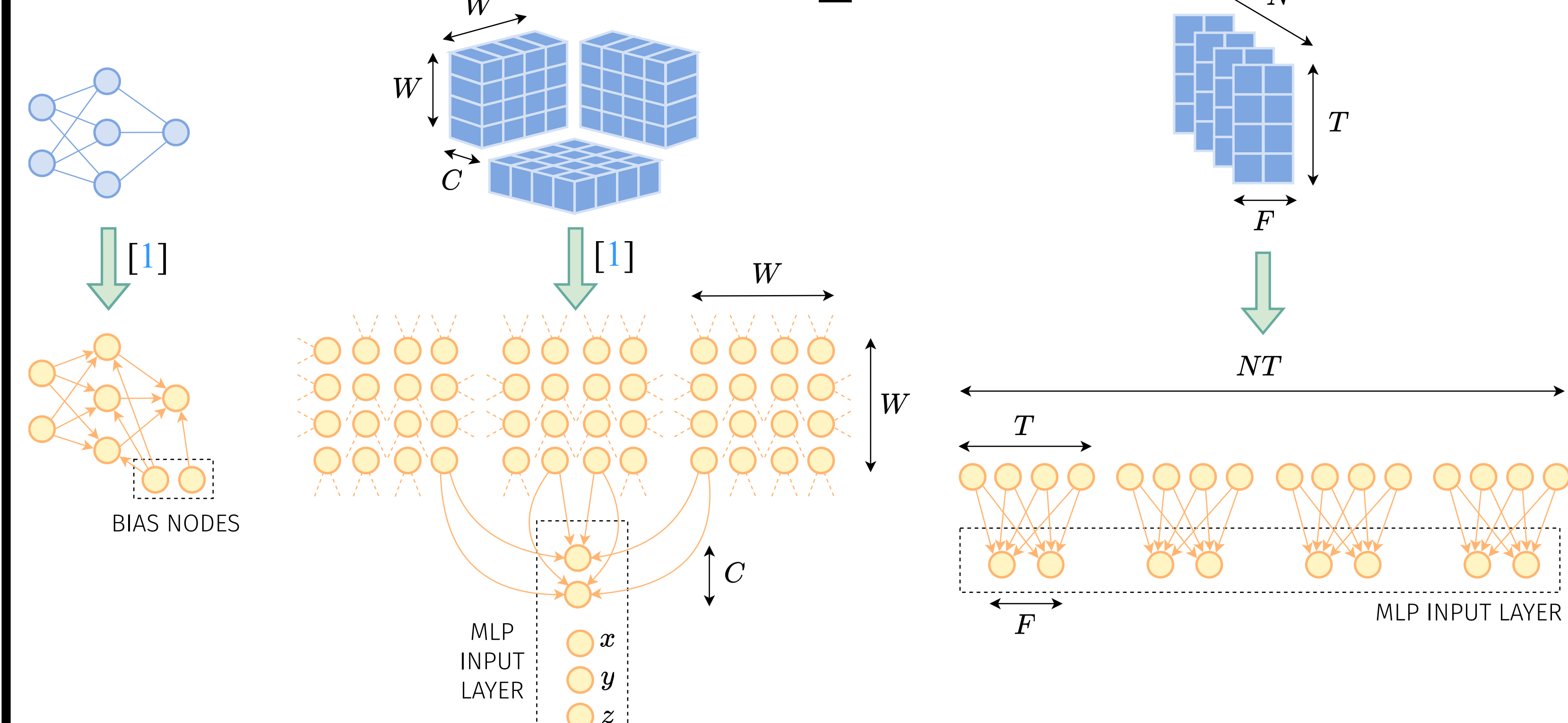
Accuracy (%) ↑											
Method	Classifier Training Set	MLP-2L	MLP-32H	TRI-2L	TRI-32H	TRI-16W	TRI-8C	HASH-2L	HASH-32H	HASH-3N	HASH-11T
nf2vec [2]	—	—	—	—	—	—	—	—	—	—	—
Cardace et al. [3]	—	—	—	—	—	—	—	—	—	—	—
$\mathcal{L}_R$ (ours)	ALL	91.3	87.4	93.2	88.3	24.8	69.4	91.9	91.2	88.3	24.5
$\mathcal{L}_{R+C}$ (ours)	ALL	85.9	83.8	87.0	84.1	72.1	30.8	89.2	87.4	86.8	27.8
$\mathcal{L}_R$ (ours)	MLP	91.3	86.1	22.6	23.2	7.9	21.0	19.7	20.6	21.8	7.6
$\mathcal{L}_{R+C}$ (ours)	MLP	86.6	81.3	63.3	43.4	59.2	13.1	51.5	50.8	55.0	24.8
$\mathcal{L}_R$ (ours)	TRI	11.9	10.6	92.1	84.8	32.1	43.6	9.2	8.8	15.0	5.2
$\mathcal{L}_{R+C}$ (ours)	TRI	60.7	58.0	86.9	83.2	63.9	30.2	59.8	61.5	66.3	28.5
$\mathcal{L}_R$ (ours)	HASH	10.7	7.4	33.9	36.3	19.1	23.0	91.6	91.4	87.8	29.2
$\mathcal{L}_{R+C}$ (ours)	HASH	47.0	40.4	40.6	34.0	43.4	19.3	89.5	87.9	86.4	25.7

Accuracy (%) ↑											
Method	Encoder Training Set	MLP-2L	MLP-32H	TRI-2L	TRI-32H	TRI-16W	TRI-8C	HASH-2L	HASH-32H	HASH-3N	HASH-11T
nf2vec [2]	—	63.7	—	—	—	—	—	—	—	—	—
Cardace et al. [3]	MLP	91.8	83.7	10.9	6.1	7.0	4.9	5.7	5.6	4.0	4.9
nf2vec [2]	—	—	—	—	—	—	—	—	—	—	—
Cardace et al. [3]	TRI	—	—	93.5	92.0	—	68.6	—	—	—	—
$\mathcal{L}_R$ (ours)	TRI	10.1	10.0	92.6	82.5	22.9	72.8	17.4	13.6	10.8	12.1
nf2vec [2]	—	—	—	—	—	—	—	—	—	—	—
Cardace et al. [3]	HASH	—	—	—	—	—	—	—	—	—	—
$\mathcal{L}_R$ (ours)	HASH	8.3	8.9	15.4	21.3	21.3	21.3	90.6	91.0	88.2	28.8

NeRFs



Graphs



Language Tasks

Brief captioning				Detailed captioning				Single-round Q&A			
Method	Test Dataset	S-BERT ↑	SimCSE ↑	Method	Test Dataset	S-BERT ↑	SimCSE ↑	Method	Test Dataset	S-BERT ↑	SimCSE ↑
LLaNA [4]				LLaNA [4]				LLaNA [4]			
—				—				—			
$\mathcal{L}_{R+C}$ (ours)				$\mathcal{L}_{R+C}$ (ours)				$\mathcal{L}_{R+C}$ (ours)			
MLP				MLP				MLP			
67.1				73.3				80.9			
TRI				TRI				TRI			
66.4				72.2				80.7			
HASH				HASH				HASH			
66.3				72.4				80.6			
Method				Method				Method			
S-BERT ↑				S-BERT ↑				S-BERT ↑			
SimCSE ↑				SimCSE ↑				SimCSE ↑			
LLaNA [4]				LLaNA [4]				LLaNA [4]			
68.6				77.4				81.0			
$\mathcal{L}_R$ (ours)				$\mathcal{L}_R$ (ours)				$\mathcal{L}_R$ (ours)			
68.3				74.1				81.6			